ALICE L WALTON SCHOOL OF MEDICINE

Hepatic Ethanol Metabolism

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Learning Objectives (2)

After completing this brick, you will be able to:

1. List the predominant enzymes, products, and coenzymes involved in the oxidative metabolism of ethanol in the liver.
2. Compare and contrast the adverse metabolic effects that excess ethanol has on the hepatic processes of gluconeogenesis and fatty acid metabolism.

CASE CONNECTION

During a clinical rotation at a family clinic, a 52-year-old man presents to the clinic with fatigue and vague right upper quadrant discomfort. He has a long history of alcohol use disorder, drinking 6–8 beers daily for the past 20 years. He reports poor dietary intake and has unintentionally lost weight over the past year. On exam, his BMI is 18 kg/m². He appears thin with mild temporal wasting. The liver edge is palpable and mildly enlarged but non-tender.

Laboratory findings:

- AST: 110 U/L
- ALT: 45 U/L
- (AST:ALT ratio >2)
- GGT: elevated
- Serum triglycerides: elevated
- Albumin: slightly low
- Total bilirubin: mildly elevated

Ultrasound demonstrates increased hepatic echogenicity consistent with fatty infiltration of the liver.

Your fellow student states, "The fatty liver findings in this patient don't really make sense. The patient is underweight and doesn't seem to consume excess calories. Could this be something like kwashiorkor, where protein deficiency prevents apolipoprotein synthesis, leading to abnormal lipid accumulation in hepatocytes? The poor diet and low albumin seem to fit that possibility."

Your attending turns to you and asks if you agree with your student colleague or not. Consider your response as you read this Brick.

GO TO CONCLUSION ↓

List the predominant enzymes, products, and coenzymes involved in the oxidative metabolism of ethanol in the liver (LO#1)

While this brick will focus on how the liver metabolizes drinking alcohol (i.e., ethanol [EtOH]) and the subsequent consequences of this metabolism, you should remember this is not the sole adverse impact of ethanol consumption. Some of the effects of alcohol ingestion, such as the psychotropic effects on the brain are directly caused by EtOH itself. In addition, EtOH directly leads to altered protein function within the GI tract as it is a potent protein denaturant and organic solvent. EtOH's chemical nature can have a spectrum of undesired impacts such as compromising absorption of dietary nutrients such as water-soluble vitamins. However, many of the acute and chronic pathophysiologic effects of alcohol relate to ethanol metabolism pathways, which are discussed below.

CLINICAL REASONING

Ethanol (CH_3CH_2OH) is the same molecule found in alcoholic beverages and common disinfectants such as hand sanitizers (e.g., GermX). Its antimicrobial effectiveness stems from its ability to form hydrogen bonds and disrupt protein structure, leading to denaturation of viral and bacterial proteins and rapid surface sterilization. However, when ingested, ethanol exerts similar protein-disrupting effects on host tissues. In the gastrointestinal (GI) tract, this can result in mucosal injury and impaired cellular function. Clinically, ethanol exposure is associated with barrier disruption (e.g., gastritis, peptic ulcers, increased intestinal permeability or "leaky gut"), alterations in the gut microbiome, and an increased risk of malignancy.

The liver metabolizes over 90% of ingested ethanol through oxidative pathways. The first step in ethanol oxidation generates acetaldehyde (an aldehyde intermediate), which, although short-lived, is highly toxic. Acetaldehyde contributes to many of the acute symptoms of alcohol consumption (e.g., hangovers) and plays a key role in the increased cancer risk associated with chronic alcohol use.

At the molecular level, acetaldehyde is highly reactive. It forms covalent adducts with DNA, lipids, and proteins, impairing their structure and function. It also promotes the generation of reactive oxygen species (ROS), further exacerbating cellular injury. Through these mechanisms, acetaldehyde drives oxidative stress, inflammation, and immune activation. Collectively, the aldehyde-mediated damage can activate Kupffer cells, resident macrophages within hepatic tissue. Immune cell activation also stimulates fibrotic pathways, making acetaldehyde a central contributor to features of alcoholic liver disease. The specific clinical aspects and progression of alcohol-associated liver disease will be explored in the Alcohol-Associated Liver Disease Brick.

Acetaldehyde formation from ethanol occurs via three distinct enzymatic systems:

- **alcohol dehydrogenase (ADH):** The major liver enzyme responsible for the first step of ethanol oxidation.
- **cytochrome P_{450} 2E1 isoform (CYP2E1):** enriched in hepatic microsomes and has a fairly high K_m for ethanol. Importantly, this select isoform is ***induced*** with chr on. As a cytochrome P450 enzyme, it contains a heme (porphyrin) prosthetic group. This is known as microsomal ethanol-oxidizing system (MEOS).
- **Catalase:** a minor peroxisomal pathway that oxidizes ethanol using hydrogen peroxide.

CLINICAL CORRELATION

Alcohol-dependent induction of CYP2E1 becomes medically significant in acetaminophen poisoning and select forms of porphyria which were covered extensively in the foundational courses (e.g. BOOM, Pharm).

In addition, the inducible microsomal ethanol-oxidizing system (MEOS, via CYP2E1) and, to a lesser extent, peroxisomal pathways contribute to the generation of reactive oxygen species (ROS) during ethanol metabolism. During periods of increased ethanol consumption, enhanced flux through these pathways leads to elevated intracellular oxidative stress. This oxidative stress is central to alcohol-associated liver injury.

Further oxidation yields relatively non-toxic acetate which can be excreted (minor route) or converted to acetyl-CoA for ATP production via the TCA cycle (major route and more clinically relevant). Acetate is considered a normal constituent of the diet so most of the harmful effects of alcohol metabolism occur upstream of acetate production itself. In contrast to the first oxidative step, this next oxidative step is catalyzed exclusively by aldehyde dehydrogenase (ALDH) enzymes. The overall reaction is depicted in **Figure 1**.

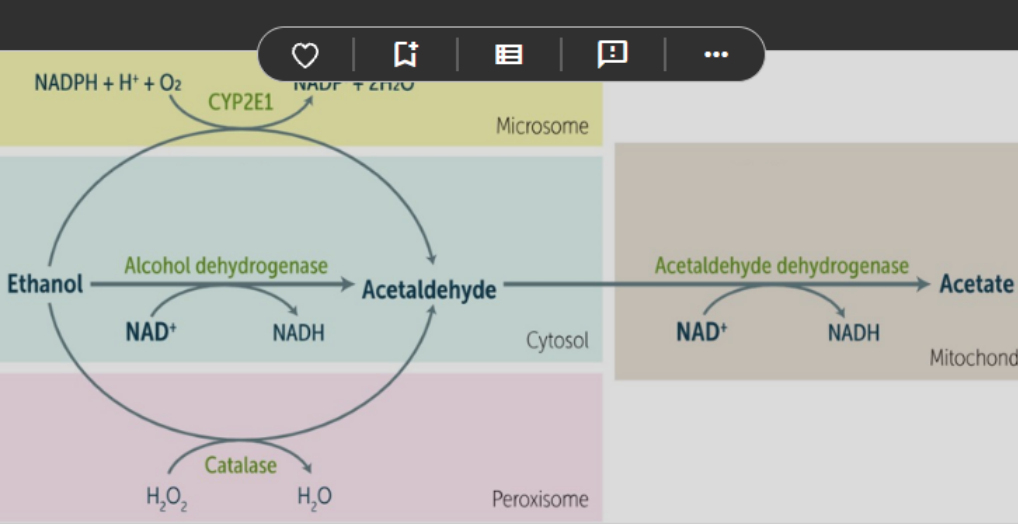


Figure 1. Reaction scheme of oxidative ethanol metabolism.

ACTIVE LEARNING QUESTION

Match the correct ethanol-metabolizing enzyme to its correct descriptor.

ADH	✓ consumes NAD⁺
CYP2E1	✓ induced by chronic or excessive alcohol intake
Catalase	✓ present in peroxisomes
ALDH	✓ common step in ethanol metabolism that yields acetate

Explanation

See figure 1 for schematic of enzymes and processes.

CLINICAL CORRELATION

Acetaldehyde, the toxic intermediate of ethanol metabolism, can be therapeutically exploited to discourage alcohol use through aversion therapy. It does not reduce the addictive cravings but induces unwanted physiological effects if alcohol is consumed. Disulfiram works by irreversibly inhibiting aldehyde dehydrogenase (ALDH) to inactivate the enzyme. As a result, when a patient consumes alcohol while taking disulfiram, acetaldehyde accumulates rapidly as it is not efficiently converted to acetate in the presence of the drug. This leads to rapid onset of unpleasant symptoms similar to *but more severe than* a hangover, including flushing, throbbing headache, copious vomiting, sweating, palpitations, dyspnea, and generalized sympathetic activation with alcohol use as little as 15-20 mL.

Genetic variation in aldehyde dehydrogenase (ALDH) enzymes contributes to individual differences in physiological responses to alcohol consumption. The *alcohol flush reaction*, which is particularly common among individuals of East Asian descent, results from reduced enzymatic activity of certain ALDH variants. These less efficient yde, leading to its accumulation. Eleva y to alcohol and commonly produce symptoms such as nausea and upper body or facial erythema (skin flushing).

While harmful, ethanol can be viewed as a source of dietary fuel as the ethanol-derived acetate can be converted to acetyl-CoA, which is primarily oxidized in skeletal and cardiac muscle. The brain and kidney moderately utilize the ethanol-derived acetate as an oxidative fuel. Transformation of acetate requires expression of the acetyl-CoA synthetase (ACS) enzyme (**Figure 2**). While the liver also expresses ACS, the acetyl-CoA derived from acetate is converted to ketone bodies due to slowing of the TCA cycle when ethanol metabolism causing NADH/NAD⁺ redox imbalance which is discussed below.

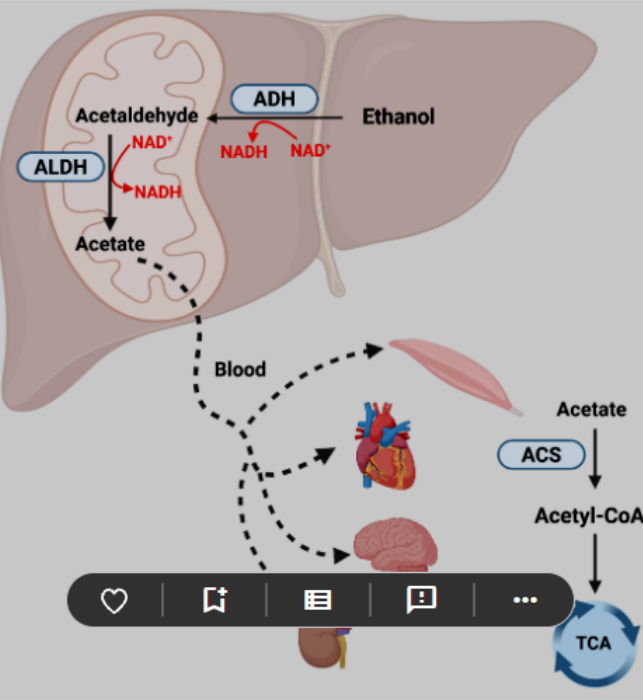


Figure 2. Oxidative catabolism of EtOH through the TCA cycle. alcohol dehydrogenase (ADH), aldehyde dehydrogenase (ALDH), acetyl-CoA synthetase (ACS)

CLINICAL REASONING

Observe how alcohol metabolism within the liver involves ***both*** the cytoplasm and mitochondria cellular compartments (Figure 2). The fact that NADH is accumulating in both the cytoplasm and mitochondria is clinically significant as metabolic pathways in both cellular locales are altered as described below.

ACTIVE LEARNING QUESTION

How does excessive ethanol metabolism alter the NADH/NAD⁺ ratio within the liver?

A. ○ decreases

B. ● increases

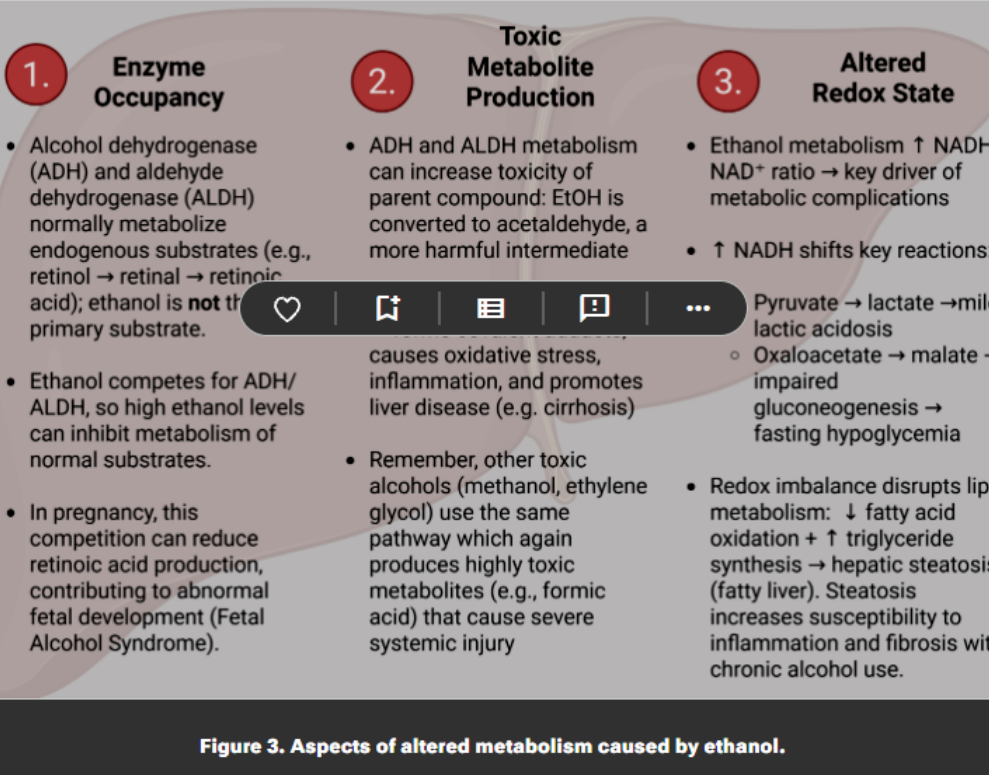
C. ○ unaltered

Explanation

(B) is correct as the liver oxidizes ethanol to acetaldehyde to acetate, it predominantly reduces the NAD⁺ coenzyme to generate NADH in the process via ADH and ALDH. If EtOH is consumed in excess, NADH will predominate and increase the NADH:NAD⁺ ratio. This more reduced environment has dire consequences which will be explored below. (A) is incorrect as this is the opposite of what occurs. (C) is incorrect as the presence of ethanol results in oxidative catabolism which reduces NAD⁺ to generate NADH.

Compare and contrast the adverse metabolic effects that excess ethanol has on the hepatic processes of gluconeogenesis and fatty acid metabolism (LO#2)

Hepatic EtOH metabolism is detrimental to health due to three main distinct but synergistic impacts (**Figure 3**). While all of these adverse impacts are clinically relevant, this brick explores 2 and 3 in detail below.


1. Enzyme Occupancy

- Alcohol dehydrogenase (ADH) and aldehyde dehydrogenase (ALDH) normally metabolize endogenous substrates (e.g., retinol → retinal → retinoic acid); ethanol is **not** th primary substrate.
- Ethanol competes for ADH/ALDH, so high ethanol levels can inhibit metabolism of normal substrates.
- In pregnancy, this competition can reduce retinoic acid production, contributing to abnormal fetal development (Fetal Alcohol Syndrome).

2. Toxic Metabolite Production

- ADH and ALDH metabolism can increase toxicity of parent compound: EtOH is converted to acetaldehyde, a more harmful intermediate causes oxidative stress, inflammation, and promotes liver disease (e.g. cirrhosis)
- Remember, other toxic alcohols (methanol, ethylene glycol) use the same pathway which again produces highly toxic metabolites (e.g., formic acid) that cause severe systemic injury

3. Altered Redox State

- Ethanol metabolism ↑ NADH/NAD⁺ ratio → key driver of metabolic complications
- ↑ NADH shifts key reactions:
  - Pyruvate → lactate →mild lactic acidosis
  - Oxaloacetate → malate → impaired gluconeogenesis → fasting hypoglycemia
- Redox imbalance disrupts lipid metabolism: ↓ fatty acid oxidation + ↑ triglyceride synthesis → hepatic steatosis (fatty liver). Steatosis increases susceptibility to inflammation and fibrosis with chronic alcohol use.

Figure 3. Aspects of altered metabolism caused by ethanol.

Key metabolic processes (e.g., glycolysis, TCA cycle, β-oxidation, etc.) are influenced by **how much NADH accumulates relative to NAD⁺**. Excess metabolism of ethanol from alcohol consumption **increases the NADH/NAD⁺ ratio** within **both the cytoplasm and mitochondria** to inhibit gluconeogenesis and fatty acid metabolism. The rest of this Brick will focus on how ethanol metabolism alters the redox balance within the liver and how this specific change disrupts gluconeogenesis and fatty acid metabolism.

Gluconeogenesis: Alcohol–associated fasting hypoglycemia

A high NADH-to-NAD⁺ ratio triggers activation of other reactions that convert NADH back to NAD⁺. Pivotal effects of this redox change favor:

- reduction of pyruvate to lactic acid
- reduction of oxaloacetate (OAA) to malate

Conversion of pyruva **cidosis** by favoring lactate formation and its conversion back to pyruvate via LDH. Conversion of OAA to malate by malate dehydrogenase dampens flux of acetyl-CoA through the TCA cycle and instead **favors ketone body production.** Elevated NADH from EtOH metabolism promotes ketone body production, and during fasting states, ketone levels can rise significantly, contributing to **metabolic acidosis**. The combination of elevated lactate with pronounced ketone body accumulation accounts for the fact that many patients develop an **anion gap metabolic acidosis.**

CLINICAL CORRELATION

Remember from previous courses, cessation of alcohol consumption is a typical dietary recommendation for gout patients. This largely stems from the fact that alcohol oxidative metabolism increases the NADH/NAD⁺ ratio favoring lactic acid production from pyruvate. Lactic acid interferes with the transporter within the kidneys needed for uric acid excretion. While the kidney reabsorbs ~90% of the uric acid, the disruption caused by the increased lactic acid is sufficient to cause hyperuricemia and a gout flare.

The most important consequence of all of these reactions is a shunting of resources (i.e., the gluconeogenic precursors of OAA and pyruvate) away from gluconeogenesis. This deprives the gluconeogenic enzymes pyruvate carboxylase and PEPCK of pyruvate and OAA, their respective substrates. Collectively, the diversion of gluconeogenic precursors into alternate pathways results in **fasting hypoglycemia** (**Figure 4**).

It is important to not n alcohol-associated hypoglycemia; therefore, hypoglycemia most commonly occurs in individuals who have depleted their hepatic glycogen stores, such as after prolonged fasting or strenuous exercise prior to alcohol consumption

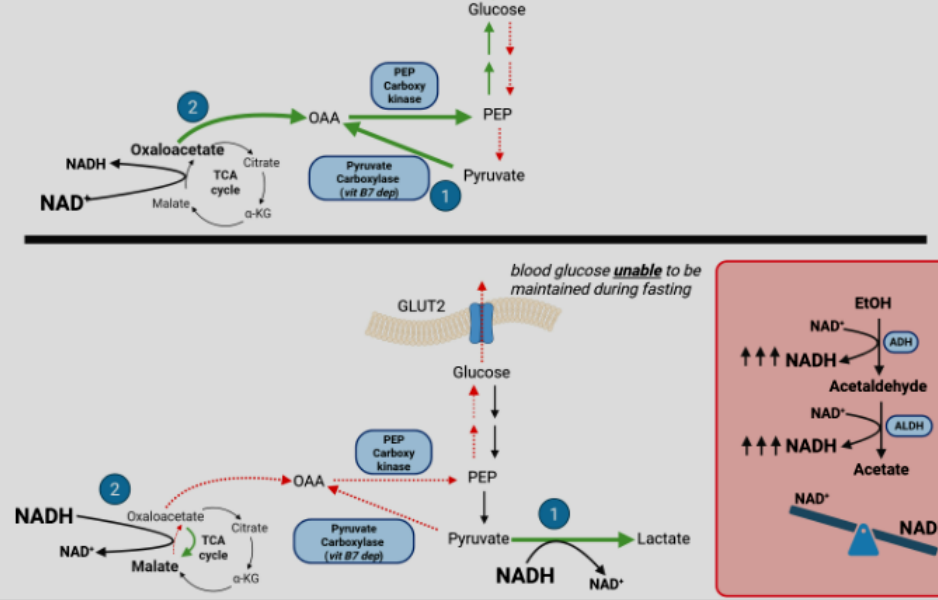


Figure 4. Metabolic changes that occur due to the altered redox state within hepatocytes during ethanol metabolism. The upper panel depicts normal hepatic gluconeogenesis that maintains blood glucose levels in the absence of ethanol. Both pyruvate and OAA supply carbon substrates for glucose synthesis (green arrows). In contrast, when ethanol is metabolized and NADH is increased, the pyruvate and OAA gluconeogenic substrates are shunted away into lactate and malate, respectively (green arrows).

CLINICAL REASONING

Thiamine Deficiency and Glucose Administration in Alcohol Use

Remember, it is potentially harmful to administer glucose alone to a hypoglycemic patient with a history of chronic alcohol use. Alcohol misuse, particularly when combined with poor nutritional intake, is commonly associated with thiamine (vitamin B1) deficiency. Thiamine is an essential cofactor for several enzymes, including the pyruvate dehydrogenase (PDH) complex, which is required to convert pyruvate to acetyl-CoA for entry into the TCA cycle. In a thiamine-deficient state, the ability to metabolize glucose is impaired. If such a patient is in a prolonged fasting or starvation state, the brain adapts by utilizing ketone bodies as an alternative fuel source. However, administration of glucose stimulates insulin release, which quickly suppresses ketogenesis and shifts metabolism back toward glucose utilization. This creates a metabolic problem in which the CNS is deprived of ketone bodies as a fuel source but is also unable to properly metabolize the administered glucose. As a result, glucose administration without prior thiamine repletion may worsen neurologic function rather than improve it. Therefore, administration of thiamine prior to glucose is essential to ensure proper oxidative metabolism and prevent neurologic deterioration. Clinically, altered mental status in such patients may be mistakenly attributed to intoxication, when it may instead reflect hypoglycemia and/or thiamine deficiency–related neurologic dysfunction.

Pyruvate and Oxaloacetate are depleted as an increase in NADH shifts pyruv alate

Fatty Acid Metabolism: Alcohol–associated hepatic steatosis

Similar to gluconeogenesis, the altered redox balance compromises proper lipid and fatty acid metabolism within hepatocytes. This is multifactorial in nature in which alcohol metabolism prevents proper fatty acid degradation, increased lipogenesis, and disrupted lipid trafficking. This dysfunction culminates in a fatty liver (i.e., hepatic steatosis) that is known to promote liver disease.

Decreased fatty acid turnover

EtOH metabolism adversely impacts the ability of hepatocytes to effectively catabolize fatty acids to acetyl-CoA via β-oxidation. Decreased fatty acid degradation partly accounts for the abnormal build-up of lipids within hepatocytes.

- Fatty acid catabolism via β-oxidation requires NAD⁺ and slows due to accumulation of NADH from ethanol oxidation. This occurs specifically at the step catalyzed by β-hydroxyacyl-CoA dehydrogenase. As a dehydrogenase enzyme (like PDH or select others in the TCA), it catalyzes oxidation/reduction reactions. To oxidize fatty acids, the enzyme must reduce NAD⁺ to NADH. With an elevated NADH:NAD⁺ ratio, oxidation of **fatty acids is not favored.**
- As the TCA cycle ailability with malate formation from OAA being favored (**see** Figure 4), the accrual of acetyl-CoA further suppresses β-oxidation at the step catalyzed by thiolase by product inhibition.

Increased fatty acid synthesis

- High levels of acetyl-CoA within hepatocytes due to EtOH metabolism slowing the TCA cycle will be diverted into lipogenesis via acetyl-CoA carboxylase.
- Increased NADH levels favor the shunting of DHAP from the glycolytic pathway into **glycerol 3-phosphate formation.** Glycerol 3-phosphate serves as the backbone of triglycerides (TG) via the esterification of three fatty acids. The synthesized TGs are subsequently packaged in VLDL particles (Figure 5).

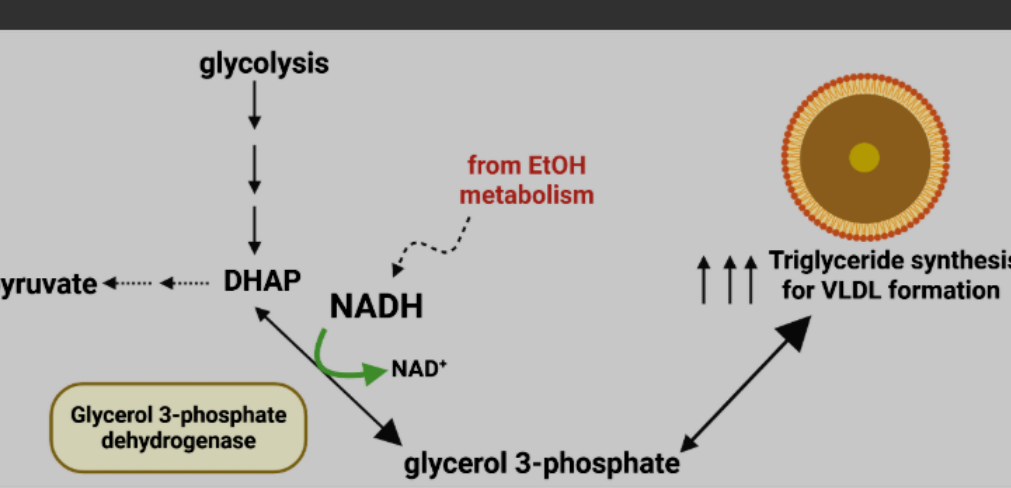

glycolysis
pyruvate
DHAP
NADH
NAD⁺
from EtOH metabolism
Glycerol 3-phosphate dehydrogenase
glycerol 3-phosphate
↑↑↑ Triglyceride synthesis for VLDL formation

Figure 5. Altered redox balance due to EtOH metabolism shift glycolytic intermediates into fatty acid synthesis pathways with subsequent VLDL packaging

Disrupted lipid trafficking

- Due to the increase in lipogenesis (see point above), **malonyl-CoA production is increased** via acetyl-CoA carboxylase. This lipid intermediate inhibits CPT-1 mediated transport of fatty acids into the mitochondrial matrix. Blocking the trafficking of fatty acids into the mitochondria further slows β-oxidation, augmenting lipid accumulation within the liver.
- The highly reactive acetaldehyde intermediate blunts lipid trafficking by disrupting normal protein function required for cellular secretion. Namely, acetaldehyde-adduct formation damages microtubules required for secretion resulting in hepatocyte retention of the very-low-density lipoprotein (VLDL) produced.

CLINICAL REASONING

The AST:ALT ratio is a useful biochemical clue in distinguishing alcohol-associated liver injury from other causes of hepatocellular damage. The classic teaching is an AST:ALT ≥ 2 strongly suggests alcohol-induced liver disease. This pattern occurs as ALT is predominately localized to the cytoplasm and AST is contained in both the cytoplasm and mitochondria. As EtOH damages both the plasma membrane and the integrity of the mitochondria via ROS and acetaldehyde, AST is typically released into the plasma to a greater extent than with other hepatic insults. In other words, the ethanol-induced mitochondrial damage disproportionately increased AST in the plasma.

In summary, EtOH metabolism creates a high-NADH state that shifts metabolism toward reduction reactions (e.g. lipogenesis) and away from oxidative pathways (e.g. β-oxidation and TCA) which has numerous health implications.

Later bricks will explore the progression of these initial features (e.g. hepatic steatosis) to alcohol-induced hepatitis and cirrhosis in further depth.

CASE CONNECTION

BACK TO INTRODUCTION ↑

You respond to your preceptor's question by stating "That's a thoughtful connection as protein deficiency can indeed impair lipoprotein export and cause fatty liver. But in this case, there's a more common explanation. As ethanol alters the NADH/NAD⁺ redox balance, it promotes lipid accumulation through a variety of mechanisms including decreased lipid turnover, increased synthesis, and disrupted trafficking of VLDL from the hepatocytes."

The intern then turns to you and asks which vitamin gives rise to NADH/NAD⁺?

You quickly reply with vitamin B3 or niacin, and somewhere in Arkansas Dr. Bridges feels warm inside and weeps a solitary tear of joy.

Summary

- Ethanol is oxidized in the liver through three enzymatic systems: alcohol dehydrogenase (rosomal ethanol-oxidizing system (MEOS via CYP2E1) as an inducible secondary pathway, and catalase as a minor peroxisomal pathway. All three systems convert ethanol to the toxic intermediate acetaldehyde, which is subsequently oxidized to acetate by aldehyde dehydrogenase (ALDH).
- The coenzyme NAD⁺ is the primary electron acceptor in ethanol oxidation. Both ADH and ALDH reduce NAD⁺ to NADH, causing a significant increase in the NADH/NAD⁺ ratio within the cytoplasm and mitochondria, respectively, during excessive ethanol metabolism.
- Acetaldehyde is a highly reactive and toxic intermediate that forms covalent adducts with DNA, proteins, and lipids, generates reactive oxygen species (ROS), activates Kupffer cells promoting fibrosis. These mechanisms underlie the development of alcoholic liver disease and contribute to increased cancer risk.
- The elevated NADH/NAD⁺ ratio impairs hepatic gluconeogenesis by diverting pyruvate toward lactate and oxaloacetate (OAA) toward malate, depleting key gluconeogenic precursors. This results in fasting hypoglycemia, lactic acidosis, and ketone body accumulation that can produce an anion gap metabolic acidosis.
- Excess ethanol metabolism promotes hepatic steatosis through three synergistic mechanisms: decreased fatty acid β-oxidation due to NADH accumulation, increased lipogenesis driven by elevated acetyl-CoA and glycerol-3-phosphate production, and disrupted VLDL trafficking caused by acetaldehyde-induced microtubule damage.

Review Questions

1. A 47-year-old man ment after being found unconscious at the bottom of a basement staircase. He has a history of alcohol use disorder and was last seen drinking heavily two hours ago. On examination, he is stuporous and has mild hyporeflexia but there is no evidence of trauma. Laboratory tests reveal: glucose 55 mg/dL, AST 125 U/L, ALT 55 U/L, bicarbonate 18 mmol/L, lactate 3.2 mmol/L, blood ethanol 120 mg/dL. Toxicology is negative for other substances. He has no previous liver disease and takes no medications. Which metabolic process most likely accounts for the patient's hypoglycemia?

A. ○ Ethanol consumption increases ketone body production to decrease the reliance on hepatic gluconeogenesis

B. ○ The altered redox state with ethanol favor lipogenesis and impedes lipid degradation

C. ○ Thiamine deficiency prevents proper oxidative metabolism of glucose via pyruvate dehydrogenase complex

D. ○ Increased NADH inhibits gluconeogenesis by shifting pyruvate to lactate

Explanation (requires correct answer)

2. A 29-year-old woman is evaluated for nausea, vomiting, and confusion after consuming large amounts of alcohol at a party, followed by prolonged fasting. She denies prior medical problems. On examination, she is confused, has moist mucous membranes, and exhibits mild tremor. Lab results: glucose 48 mg/dL, AST 58 U/L, ALT 40 U/L, bicarbonate 19 mmol/L, lactate 2.7 mmol/L, blood ethanol 80 mg/dL. No evidence of hepatic failure or icterus. Urine toxicology is negative. Which metabolic intermediate accumulates in he bolism, contributing to this patient's findings?

A. ○ acetaldehyde

B. ○ lactate

C. ○ NADH

D. ○ NADPH

E. ○ Pyruvate

Explanation (requires correct answer)

3. A 42-year-old man presents to his primary care physician with fatigue and mild upper abdominal discomfort that has persisted for the past two weeks. He reports drinking at least 6 beers a day for the last 8 years but denies illicit drug use. His diet is poor as he routinely eats one small protein-heavy meal a day (e.g. a can of tuna). He does not take any medications and smokes occasionally. His medical history is otherwise unremarkable. On exam, the patient appears well and is borderline underweight for his height. There is mild tenderness to palpation over the right upper quadrant but no hepatomegaly is noted. An abdominal ultrasound revealed increased hepatic echogenicity consistent with hepatic steatosis without evidence of masses or cirrhosis. Which of the following most likely accounts for the patient's findings regarding disrupted metabolism?

A. ○ enzymatic activation of HMG-CoA reductase

B. ○ impaired release of VLDL particles from hepatocytes

C. ○ sustained insulin signaling promoting acetyl-CoA formation from glucose

D. ○ malabsorption of dietary fat due to the presence of ethanol in the dietary tract

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ List the predominant enzymes, products, and coenzymes involved in the oxidative metabolism of ethanol in the liver.

02 ☐ Compare and contrast the adverse metabolic effects that excess ethanol has on the hepatic processes of gluconeogenesis and fatty acid metabolism.

Objective Confidence: 0/2

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